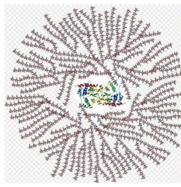
Glycogen Storage Diseases

GLYCOGEN

- largest amount of glycogen is found in skeletal muscle
- highest concentration of glycogen is in the liver
- stored in cytosolic granules mostly in the liver + muscle
- enzyme for glycogen degradation + synthesis are in the cytosol close to the glycogen granules.
- some glycogen degradation also happens in the lysosomes. (via acid maltase)

glycogen metabolism



α(1-4) linkages in glycogen filaments
α(1-6) linkages form the branch points

- Branched structure → allows for many molecules of regulated enzymes of synthesis + degradation to act on glycogen.

glycogen metabolism is tightly regulated so synthesis + degradation do not occur at the same time.

Purpose of glycogen metabolism

Rapid synthesis of glycogen for storage of glucose when glucose levels are high

Rapid degradation when glucose 6-P is needed.

① release of free glucose in the blood
② usage in glycolysis (muscle cells) to generate ATP.

Glycogen Synthesis

- Glycogen synthesis is activated by insulin
- glycogen synthase is the regulated enzyme and is active when it is dephosphorylated.

Insulin ⟶ ⊕ activates protein phosphatase-1 which cleaves the phosphates that were added by PKA.

- glycogen synthesis is inhibited by the cAMP system (glucagon/epinephrine)
- glucagon + epinephrine activate protein kinase A which then phosphorylates the key regulatory enzymes
  ⊖ deactivates glycogen synthase
  (inactive in phosphorylated state)

Glycogen Degradation

Liver glycogen Degradation

- always needs hormonal activation by glucagon or epinephrine.
- Degradation only during fasting + stress situations.

Muscle glycogen Degradation

- activated by calcium binding during contraction
- In flight/fight situations, epinephrine accelerates muscle glycogen degradation.

Glycogen
↓ Glycogen phosphorylase phosphorolytically forms Glucose 1-P until 4-5 residues before a branch point
Limit Dextrin is formed which has short branches
↓ Debranching enzyme - has 2 enzyme activities: 4:4 transferase (generates longer branches) + 1:6 glucosidase (releases free glucose)
Glycogen
↓ Glycogen Phosphorylase continues degradation until limit dextrin is formed again.

Liver glycogen metabolism + muscle glycogen metabolism have different purposes.

Liver: purpose = release glucose into the blood for other cells.

Hepatic glycogenolysis = under strict control of glucagon + epinephrine. + takes place only during fight/flight situations.

Skeletal muscle purpose = generate glucose 6-P for glycolysis + energy metabolism for the muscle.

- Independent of hormonal control unless flight/flight
- linked to Ca^{2+} release via contraction of muscle
- glycogen degradation is optimized by epinephrine.

GSD Type	Deficient enzyme	Glycogen amount, structure and other	Organ affected, characteristics Blood glucose levels
I Von Gierke disease	Glucose 6-phosphatase (bound in ER, Type Ia) Or glucose 6-P translocase (Type Ib)	High glycogen amount in the liver and the kidney. (can't break down glycogen) Normal glycogen structure.	Liver: severe hypoglycemia, hepatomegaly and kidney disease. Early death if untreated. Treatable by uncooked corn starch meal or nocturnal gastric glucose infusion.
II Pompe disease	Lysosomal (acid) glucosidase (acid maltase)	Normal glycogen amount and normal glycogen structure in cytosol, + normal blood glucose. High amount of glycogen in lysosomes leading to the lysosomal rupturing in severe cases.	Heart, liver, muscle: early death by heart failure. in infantile form Massive cardiomegaly, Myopathy, hypotonia, + weakness Treatable with enzyme infusion. Normal blood glucose levels. Infantile (most severe) and late-onset forms. - early onset → early death due to heart failure, Late onset = breathing probs.
III Cori disease	Glycogen debranching enzyme (4:4 transferase)	High amount of glycogen in liver, heart and muscle. (glycogen can't be broken down) Abnormal glycogen structure with short branches (limit dextrinosis). (an intermediate structure accumulates)	Liver: mild hypoglycemia and hepatomegaly. Muscle, heart: weakness, hypotonia and cardiomyopathy. Muscular dystrophy.
IV Andersen disease	Glycogen branching enzyme (4:6 transferase)	Low amount of glycogen in liver and muscle. (glycogen can't be formed + stored) ↓ less branches Abnormal glycogen structure with long unbranched glucose chains (attack by immune system causing scarring).	Liver: infantile cirrhosis. (due to immune sys. attack) Hepatomegaly due to cirrhosis. Muscle: infantile hypotonia. Early death.
V McArdle syndrome	Muscle glycogen phosphorylase	High amount of glycogen only in muscle. (hepatic isozyme = normal) Normal glycogen structure.	Muscle: weakness and cramping due to lack of ATP. No increase of lactate in blood after exercise as glycogen degradation is deficient (normally lac. would ↑) Normal blood glucose levels. serum CK-MM ↑ed s/p forced exercise
VI Hers disease	Liver glycogen phosphorylase	High amount of glycogen only in liver. (muscle isozyme = normal) Normal glycogen structure.	Liver: [illegible] hypoglycemia. Hepatomegaly, growth retardation.
VII Tarui disease	Glycolysis: Muscle PFK-1 RBC PFK-1 (M isoform)	High glycogen amount in muscle. Normal glycogen structure. Deficient glycolysis in muscle and RBC.	Muscle: lack of ATP leads to muscle cramping, no rise in lactate. RBC: lack of ATP leads to hemolysis. Normal blood glucose levels.

NOTES

- Type Ia vs Ib: Ia ↓ enzyme itself; Ib ↓ transporter
- Categorized as: glycogen storage, lysosomal storage, and muscular disease - progressive neuromuscular disease
- pts may need a wheelchair by the age of 50-60 yrs. ∅ scarring
- scarring
- disease often goes undxed b/c child just seems to be tired + unmotivated. - Pts can show rhabdomyolysis after forced exercise due to lack of ATP.
- Hemolysis occurs in this disease due to 50% PFK1 deficiency in RBCs